RAMDE - Engenharia Orientada a Requisitos e Modelos

Mestrado em Engenharia de Sistemas Computacionais Críticos

Lecture 03

Project Statement

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Course Project: Goals

The goal of the project is to apply the fundamentals and knowledge from the lectures into a proposed problem/domain that requires modeling, evaluation and code generation.

The Process

- Select a topic/problem for your project (see last slides for ideas);
- Lectures will provide fundamentals and concepts of Model Driven Engineering and other topics of the course;
- Lab Class will provide examples of the project concerns (using the light switch example) that you must first replicate and then develop (and extend) for your own problem/domain;
- The Project Concerns are:
  - Modeling Requirements
  - Modeling Use Cases
  - Modeling Domain
  - Modeling State Machines
  - Contraints for Models
  - Traceability
  - Code Generation

Project Concern Goals

- For each concern a demonstration will be made; You should replicate the demonstration in your repository;
- Specific goals will be presented for each concern;
- You should adapt/extend the demonstration code to achieve the goals in the context of your problem/domain and add this to the repository;
- For each concern produce a small report in the repository, using markdown, documenting your work; this should include
  - Analysis of the problem (e.g., explain requirements)
  - Design of the solution (e.g., explain meaningful solution details)
  - Tests of your solution (e.g., you may use models from your domain)
  - Discuss alternatives (maybe showing design and implementation details)

Ideas for the Project Problem/Domain

- You should search for possible examples of problems in bibliography, such as in the book:
- "**A Practical Guide to SysML, The Systems Modeling Language**"
  - Sanford Friedenthal, Alan Moore, and Rick Steiner. 2014. A Practical Guide to SysML, Third Edition: The Systems Modeling Language (3rd. ed.). Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
- When in doubt, check with the teacher if the problem is feasible/acceptable
- The last slides present some examples extracted from the previous book

Ideas for the Project Problem/Domain: Constraints

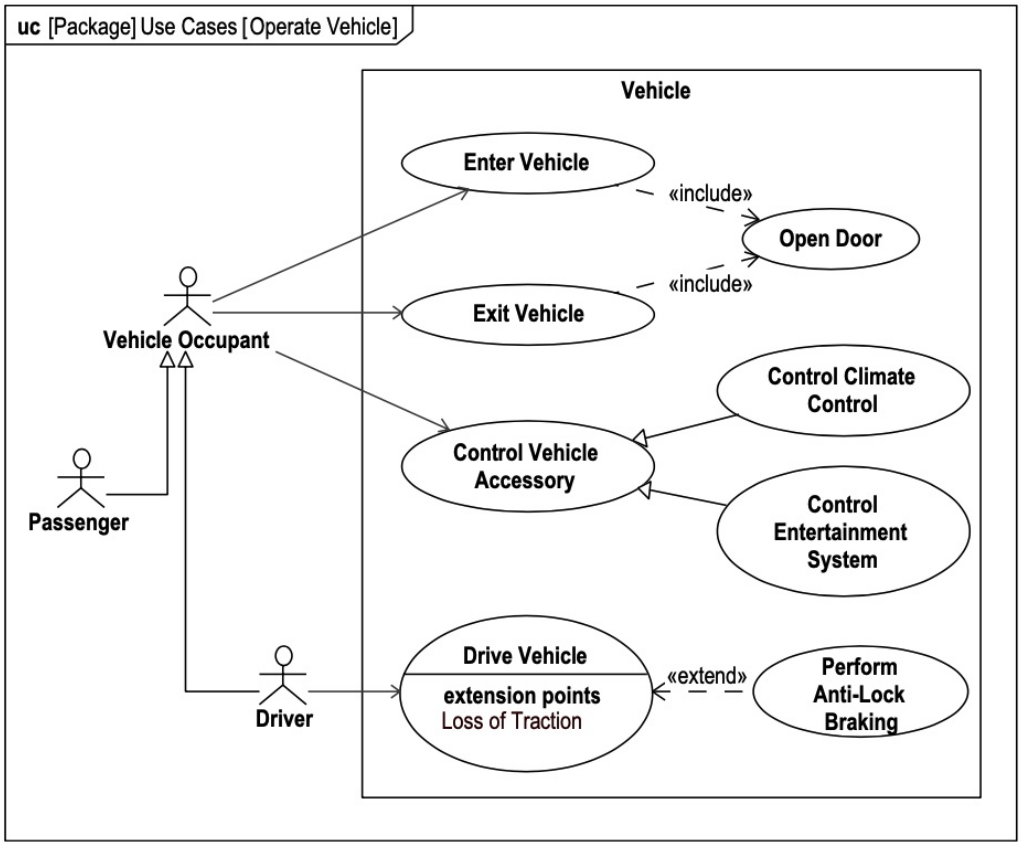
Your problem should

- be modeled with a small number of parts
- include at least one part/component that has behaviour that requires state machines
- require a "rich" set of domain rules that must be verified or tested
- the model should be complete enough to generate runnable code that dmonstrates the majority of the requirements

Idea 1: Automobile System

Chapter 4: **Design an Automobile System**

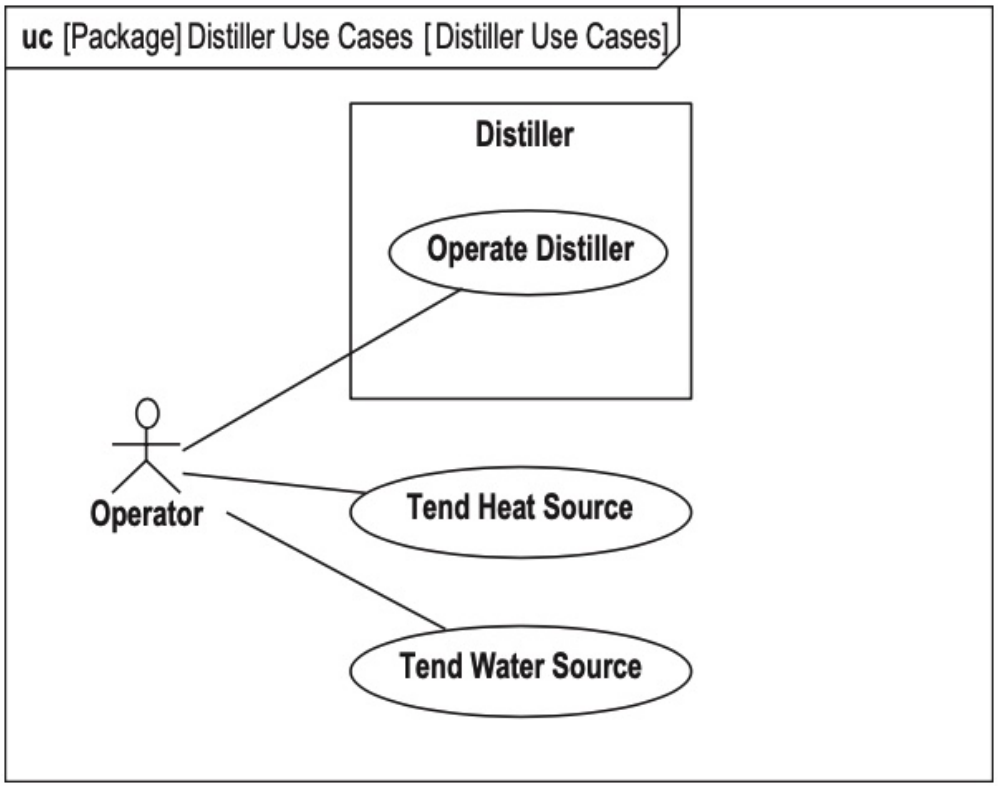
- "In this simplified example, selected aspects of the design are considered to support an initial trade-off analysis. The trade-off analysis includes an evaluation of alternative vehicle configurations that included a 4-cylinder engine and a 6-cylinder engine to determine if they can satisfy the acceleration and fuel efficiency requirement."



Idea 2: Water Distiller

Chapter 16: **Water Distiller**

- "This chapter contains an example that describes the application of SysML to the design of a water distiller intended for use in remote, undeveloped areas of the world."



Idea 3: Residential Security System

Chapter 17: **Residential Security System**

- "The example in this chapter describes the application of SysML to the development of a residential security system using the Object-Oriented Systems Engineering Method (OOSEM)"

